

OSCAT

Open Source Community for Automation
Technology

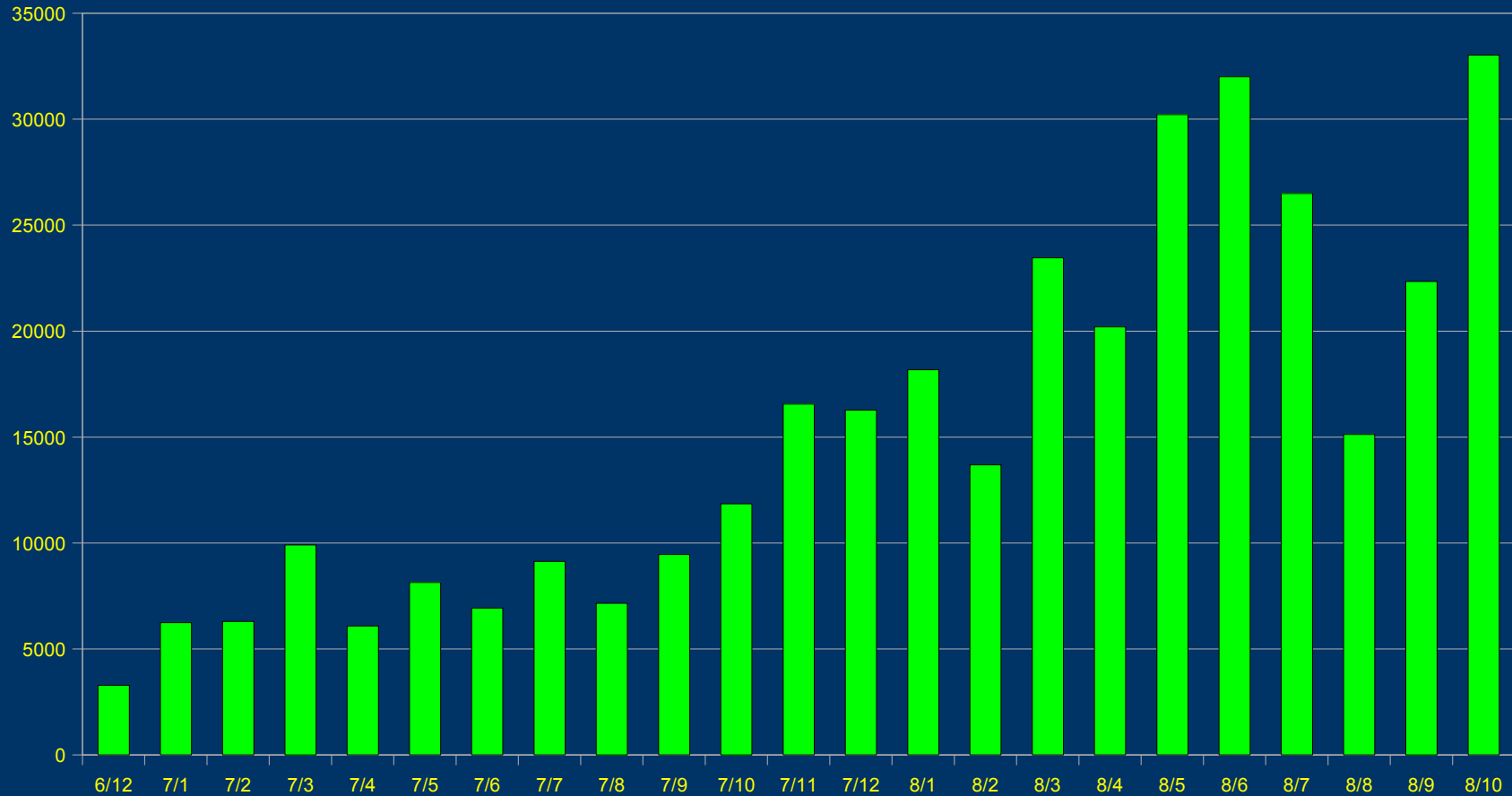
Nov 2008

OSCAT History

- Aug. 2006 Library 1.0 for Codesys 2.3
- Dec. 2006 Support Forum opened
- Feb. 2007 Library 1.4 for STEP7
- Feb. 2007 Documentation in German
- Sep. 2007 Library 2.0
- Jan. 2008 Library for PC Worx Multiprog
- Apr. 2008 added Support for Codesys 30
- Oct. 2008 increased compatibility

- Current version 3.01 (Nov. 2008)

OSCAT forum Page hits



Open Source Library

- IEC61131-3 conform
- Easy transfer between different systems
- Source code of all modules is open
- Tested on many different Hardware platforms
- Tested in many different applications
- Highly optimized code for optimal performance
- Continuous maintenance
- Monthly releases and updates
- Detailed revision and release history
- Revision tracking on module level



Cross Platform availability

- Codesys 2.3
- Codesys 3.0
- Multiprog / PC-Worx
- STEP7
- Text format for easy import in other systems
- Only Standard Library needed
- Based on IEC61131-3 functions

Most complete Library

- The OSCAT Library keeps growing
- Continuous improvement of code
- Inputs from many users and applications
- Open source guarantees flexibility
- OSCAT is optimized for 32 bit non FPU CPUs
- Extraordinary performance also on 16Bit CPUs

Planned activities

- Object oriented library
- Extensive networking functions
- TCP/IP and UDP stacks
- Continuous improvement of performance
- Continuous maintenance and service releases
- Data logging over TCP/IP functions with JAVA code for Servers

Testing and Verification

- Development on different systems assures independence of code
- Runtime tests on different hardware systems
- Testing with random patterns
- Test of math functions to 6 digits resolution
- Test program for all functions
- Test projects verify the library on different systems
- Verification under different development systems
- OSCAT works together with most other libraries

Buffer Management

- Modules to handle large arrays of Byte
- Initialization and clear
_BUFFER_INIT, _BUFFER_CLEAR
- Insert and retrieve from Buffer
_BUFFER_INSERT, _STRING_TO_BUFFER, BUFFER_TO_STRING
- Searching
BUFFER_SEARCH



List Management

- Modules to handle lists
- Lists are multiple elements separated by a separation character stored in STRING(255)
- Insert and retrieve from list
- Evaluate length of list LIST_LEN
- Clean list from empty elements



Building Automation

- Actuators for Valves, Flaps, Pumps...
- Electrical, Switches, Dimmers, Timers...
- HVAC
- Blind Control



Device Drivers

- Interfaces to Standard Devices
- IRTRANS Infrared Transceivers over TCP/IP
- Decoder for DCF77 receiver

Automation

- Incremental decoder with quad resolution
- Profile generation
- Drivers
- Signal generators
- Interlock
- Tuning
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Control

- Basic building blocks
Integrator, Differentiator, LT1, LT2, Delay...
- Advanced building blocks FT_PI, FT_PDT1 ...
- Anti Wind-Up circuitry
- Modular controller design and building blocks
- Regulators PID, PI, Hysteretic
- Control set calculation for PI and PID
- Signal Filters

Conversion

- Converting between different Units:
- Temperature
- Pressure
- Length
- Speed
- Compass directions
- Wind Speeds
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Measurement

- Calibrate, Offset, Scale
- Bar graph Lin or Log
- Cycle time, ontime
- Time and Pulse measurement
- Ontime with 64 Bit resolution
- Consumption Meter and Statistics

System Time Base

- T_PLC_MS() and T_PLC_US()
- Standard interface for reading the system timer
- Millisecond and microsecond resolution
- Debug functions for overflow
- Debug functions for speed increase to test slow processes.
- TC_MS, TC_US, TC_S returns cycle time in microseconds, milliseconds or seconds

Sensor Interface

- PT, NI, SI Sensor formulas
- Calculate temperature from resistance
- Calculate resistance from temperature
- Scale and offset calibration
- Compensate for parasitic resistors
- Redundant Analog or Digital sensor interfaces

Signal Generators

- Standard functions
SIN, triangle, ramp, square wave
- Adjustable offset, delay and amplitude
- Pulse and Bitmap generators
- Modulated generators, PWM
- Controlled Ramp Functions
- Random time and pulse generators
- Time and Logic Sequencers

Signal Processing

- Transfer functions for analog signals
- Input and output adjustments for A/D and D/A
- Sample and hold circuits
- Multiplexing and Fading
- Signal Delay
- Offset, Override and Mixing Functions
- Scaling for byte and boolean inputs
- Staircase functions

Boolean Logic

- Flip flops, latches and shift registers
- Edge and ramp triggered logic
- CRC and parity functions
- Matrix keyboard scanner
- Stack and fifo memory
- TP_R Pulse generator retriggerable
- B_TRIG, DTRIG Trigger any edge or DWORD
- Programmable pulse generators or sequencers
- Conversion BCD, DWORD, BYTE, HEX



Networking

- Support for TCP/IP and UDP
- Basic functions for protocol stacks and communication
- SNTP simple network time protocol
- IPv4 address decoder
- DNS resolution
- HTML coding and decoding
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ESR

- Error and Status Reporting
- Standardized interface to OSCAT building blocks for data logging, debugging and status reporting
- Event monitoring
- Collection of data and status of the process
- Time stamp and address label
- Interface to servers via TCP/IP
- Data and activity logging over TCP/IP

Mathematics

- Extensive mathematical functions over 100 mathematical functions
- Functions for complex numbers
- Functions for vector analysis
- Linear and polynom interpolation
- Geometry functions
- Array handling
- Statistical functions
- Double precision mathematics



STRING

- Extensive string functions
over 60 string functions
- Conversion functions
- Uppercase and Lowercase
- Support for extended ASCII > 127 (Ä,Ö,Ü)
- Enhanced FIND and Replace
- Trim functions
- String length up to 255 characters
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TIME & DATE

- Extensive Time and Date functions
- RTC Blocks with millisecond resolution
- DCF77 receiver and decoder
- Date functions with adaption for STEP7
- Sun position, sun rise and sun set calculations
- Easter and holiday calculations
- Complete set of Date and Time calculations
- Structured time format and functions